Masked Trajectory Models for Prediction, Representation, and Control

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Wu, Philipp, Arjun Majumdar, Kevin Stone, Yixin Lin, Igor Mordatch, Pieter Abbeel, and Aravind Rajeswaran. "Masked Trajectory Models for Prediction, Representation, and Control." ICML 2023

Introduction

This paper introduces Masked Trajectory Models (MTM) as a generic abstraction for sequential decision making.

- Self-supervised learning: takes a trajectory and aims to reconstruct the trajectory
- Training method: randomized masking
- Versatile capabilities: forward dynamics model, inverse dynamics model, offline RL agent

Motivation

Previous paradigms necessitate integration of multiple components to obtain decision-making policies:

- Model-based RL: world models
- Actor-critic methods: critics
- Stabilization techniques

Goal: borrow the success from self-supervised learning (masked prediction) and transformer sequence models, to create a generic and versatile models

Method

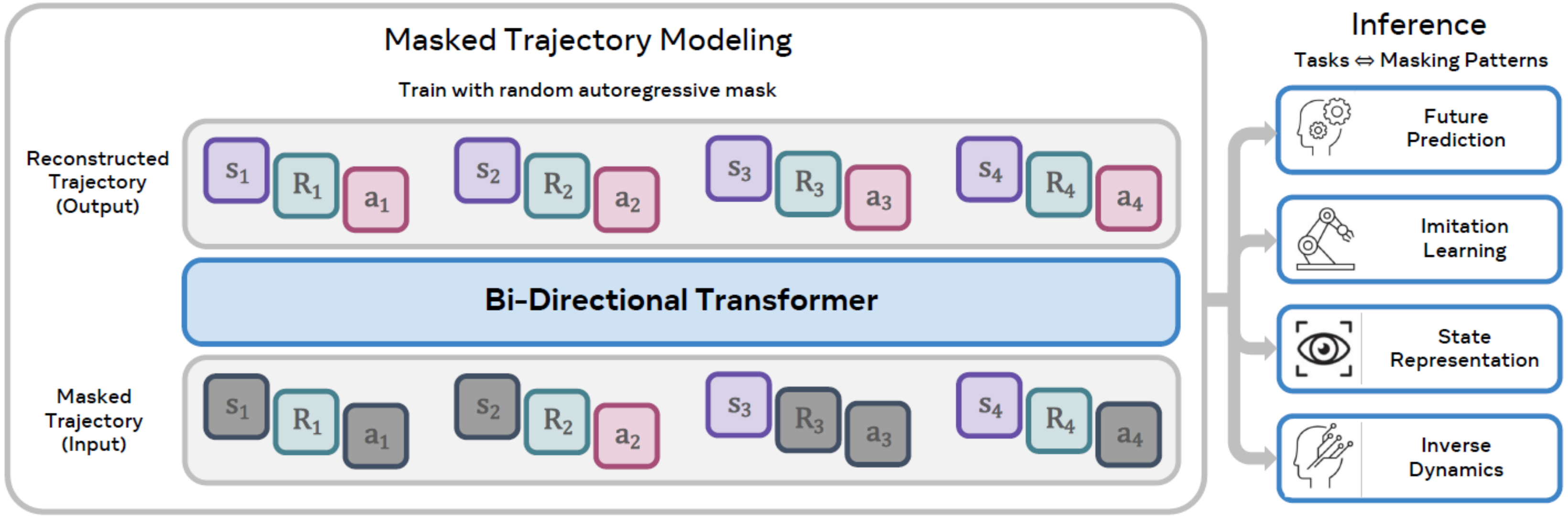


Figure 1. **Masked Trajectory Modeling (MTM) Framework.** (Left) The training process involves reconstructing trajectory segments from a randomly masked view of the same. (Right) After training, MTM can enable several downstream use-cases by simply changing the masking pattern at inference time. See Section 3 for discussion on training and inference masking patterns.

Trajectory $\tau := (s_k, a_k, s_{k+1}, a_{k+1}, \cdots, s_t, a_t)$

Reconstruction $\hat{\tau} = h_\theta(\text{Masked}(\tau))$

- $\text{Masked}(\tau)$: $(s_k, \_, \_, a_{k+1}, \cdots, s_t, \_)$ h_θ: MTM

- One Model, Many Capabilities
- Heteromodality
- Data Efficiency
- Representation Learning

Method

Trajectory with M different modalities:

$$\tau = \{(x_1^1, x_1^2, \cdots, x_1^M), \cdots, (x_T^1, x_T^2, \cdots, x_T^M)\}$$

Modality-specific encoders:

$$z_t^m = E_\theta^m(x_t^m), \forall\, t \in [1, T], m \in [1, M]$$

Embeded/tokenized trajectory:

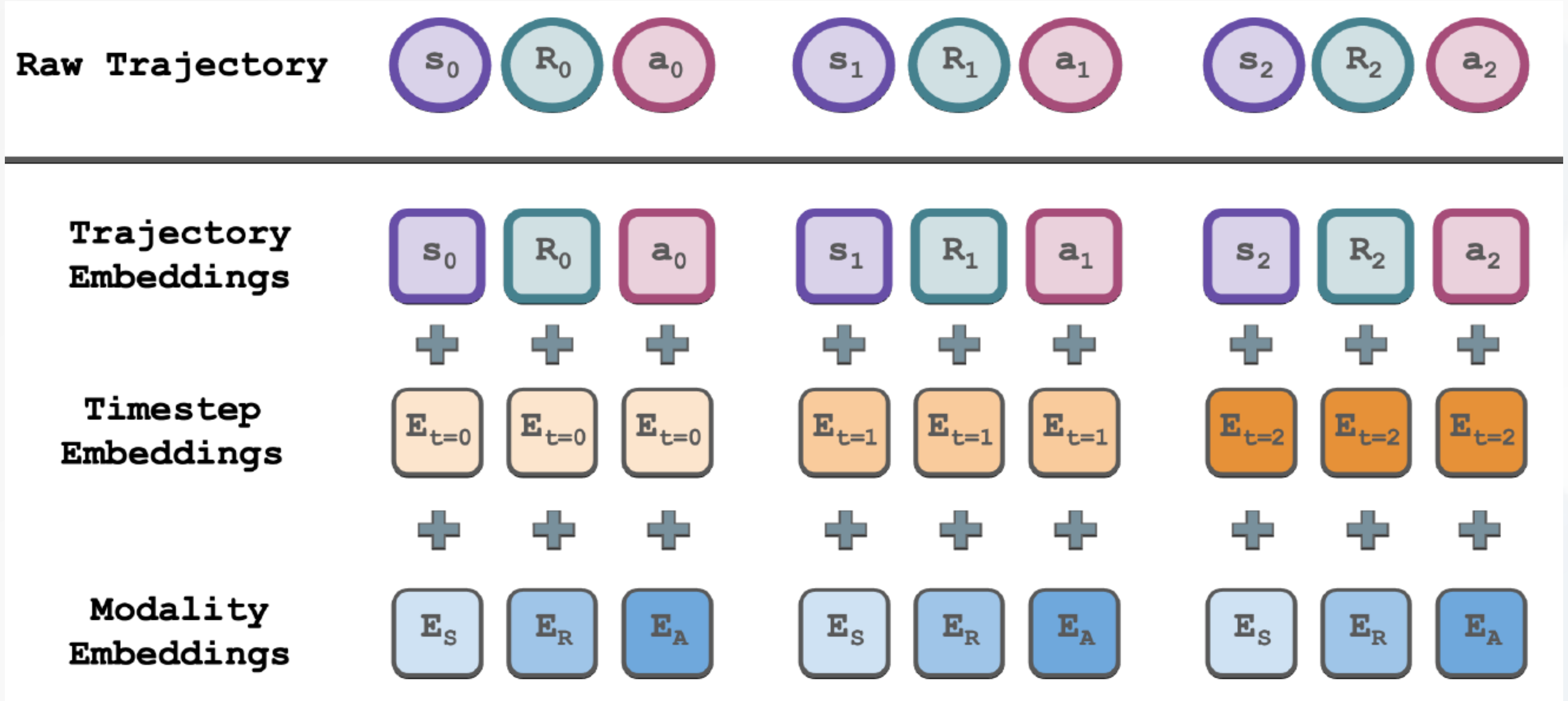
$$\tau = (z_1^1, z_1^2, \cdots, z_1^M, \cdots, z_t^m, \cdots, z_T^M)$$

Learning objective:

$$\max_\theta \mathbb{E}_\tau \sum_{t=1}^{T} \sum_{m=1}^{M} \log P_\theta(z_t^m | \text{Masked}(\tau))$$

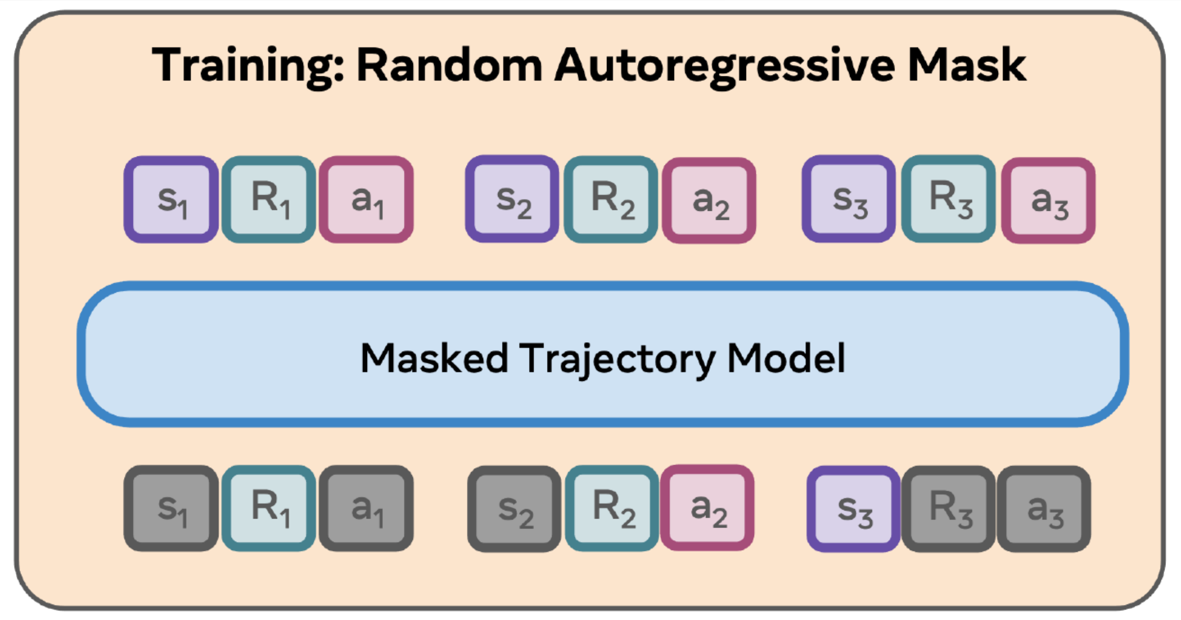
Method

Embeddings: additionally add timestep embeddings and modality type embeddings.



Method

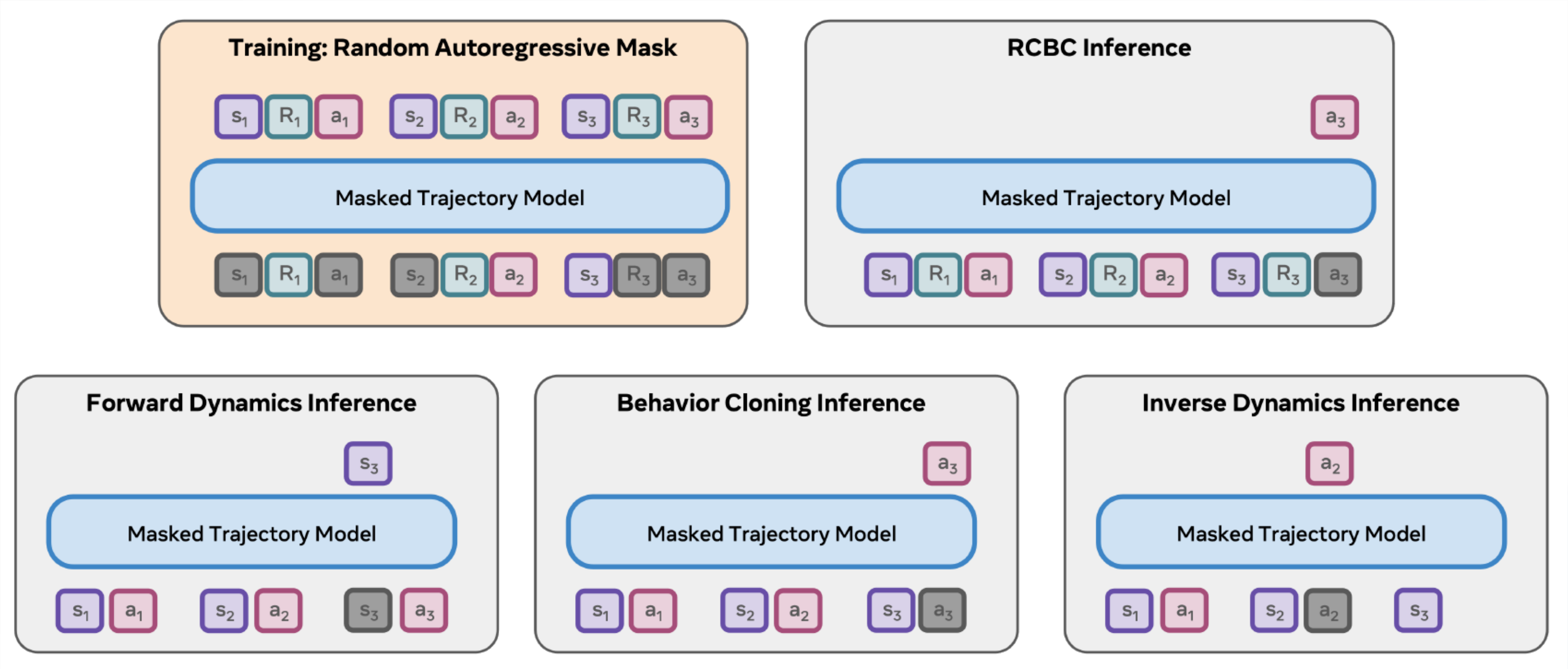
Masking Pattern: random autoregressive masking.



At least one masked token must have no future unmasked tokens. This means the last element in the sequence must necessarily be masked.

Method

Versatility: Once trained, the MTM network can take on different roles by simply using different masking patterns at inference time.



Experiments

Offline RL results

Table 1. **Results on D4RL.** Offline RL results on the V2 locomotion suite of D4RL are reported here, specified by the normalized score as described in Fu et al. (2020). We find that MTM outperforms RvS and DT, which also use RCBC for offline RL.

Environment	Dataset	BC	CQL	IQL	TT	MOPO	RsV	DT	**MTM (Ours)**
HalfCheetah	Medium-Replay	36.6	45.5	44.2	41.9	42.3	38.0	36.6	43.0
Hopper	Medium-Replay	18.1	95.0	94.7	91.5	28.0	73.5	82.7	92.9
Walker2d	Medium-Replay	26.0	77.2	73.9	82.6	17.8	60.6	66.6	77.3
HalfCheetah	Medium	42.6	44.0	47.4	46.9	53.1	41.6	42.0	43.6
Hopper	Medium	52.9	58.5	66.3	61.1	67.5	60.2	67.6	64.1
Walker2d	Medium	75.3	72.5	78.3	79.0	39.0	71.7	74.0	70.4
HalfCheetah	Medium-Expert	55.2	91.6	86.7	95.0	63.7	92.2	86.8	94.7
Hopper	Medium-Expert	52.5	105.4	91.5	110.0	23.7	101.7	107.6	112.4
Walker2d	Medium-Expert	107.5	108.8	109.6	101.9	44.6	106.0	108.1	110.2
Average		51.9	77.6	77.0	78.9	42.2	71.7	74.7	78.7

MTM uses Return Conditioned Behavior Cloning (RCBC) mask

Experiments

MTM Capabilities

Table 2. **Evaluation of various MTM capabilities.** MTM refers to the model trained with the random autoregressive mask, and evaluated using the appropriate mask at inference time. S-MTM ("Specialized") refers to the model that uses the appropriate mask both during training and inference time. We also compare with a specialized MLP baseline trained separately for each capability. Note that higher is better for BC and RCBC, while lower is better for FD and ID. We find that MTM is often comparable or better than training on specialized masking patterns, or training specialized MLPs. We use a box outline to indicate that a single model was used for all the evaluations within it. The right most column indicates if MTM is comparable or better than S-MTM, and we find this to be true in most cases.

Domain	Dataset	Task	MLP	S-MTM (Ours)	MTM (Ours)	(MTM) $\gtrsim$ (S-MTM)?
D4RL Hopper	Expert	(↑) BC	111.14 ± 0.33	111.81 ± 0.18	107.35 ± 7.77	✓
	Expert	(↑) RCBC	111.17 ± 0.56	112.64 ± 0.47	112.49 ± 0.37	✓
	Expert	(↓) ID	0.009 ± 0.000	0.013 ± 0.000	0.050 ± 0.026	✗
	Expert	(↓) FD	0.072 ± 0.000	0.517 ± 0.025	0.088 ± 0.049	✓
D4RL Hopper	Medium Replay	(↑) BC	35.63 ± 6.27	36.17 ± 4.09	29.46 ± 6.74	✗
	Medium Replay	(↑) RCBC	88.61 ± 1.68	93.30 ± 0.33	92.95 ± 1.51	✓
	Medium Replay	(↓) ID	0.240 ± 0.028	0.219 ± 0.008	0.534 ± 0.009	✗
	Medium Replay	(↓) FD	2.179 ± 0.052	3.310 ± 0.425	0.493 ± 0.030	✓
Adroit Pen	Expert	(↑) BC	62.75 ± 1.43	66.28 ± 3.28	61.25 ± 5.06	✓
	Expert	(↑) RCBC	68.41 ± 2.27	66.29 ± 1.39	64.81 ± 1.70	✓
	Expert	(↓) ID	0.128 ± 0.001	0.155 ± 0.001	0.331 ± 0.049	✗
	Expert	(↓) FD	0.048 ± 0.002	0.360 ± 0.020	0.321 ± 0.048	✓
Adroit Pen	Medium Replay	(↑) BC	33.73 ± 1.00	54.84 ± 5.08	47.10 ± 7.13	✗
	Medium Replay	(↑) RCBC	41.26 ± 4.99	57.50 ± 3.76	58.76 ± 5.63	✓
	Medium Replay	(↓) ID	0.308 ± 0.004	0.238 ± 0.004	0.410 ± 0.064	✓
	Medium Replay	(↓) FD	0.657 ± 0.023	0.915 ± 0.007	0.925 ± 0.026	✓

S-MTM, trains a specialized model for each capability using the corresponding masking pattern at train time.

Behavior Cloning (BC): Predict next action given state-action history.

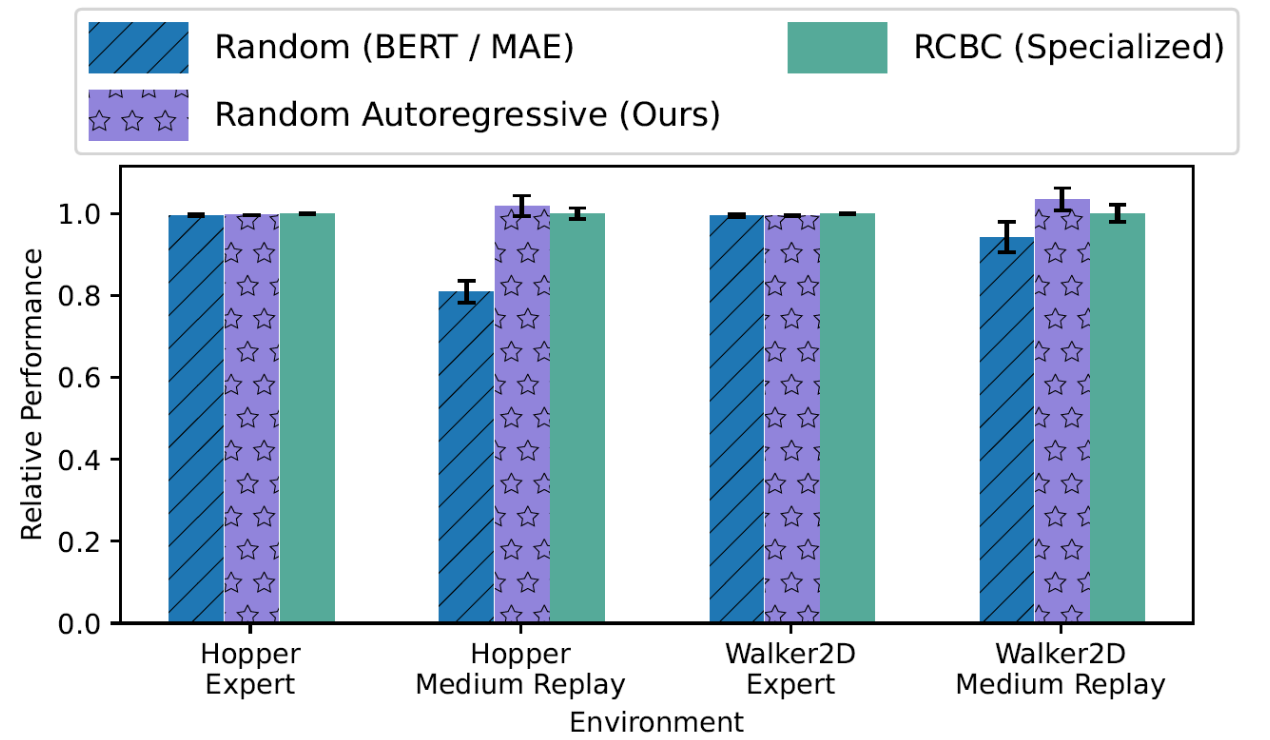
Return Conditioned Behavior Cloning (RCBC): similar to BC, additionally conditions on the desired Return-to-Go.

Inverse Dynamics (ID): predict the action using the current and future desired state.

Forward Dynamics (FD): predict the next state given history and current action.

Experiments

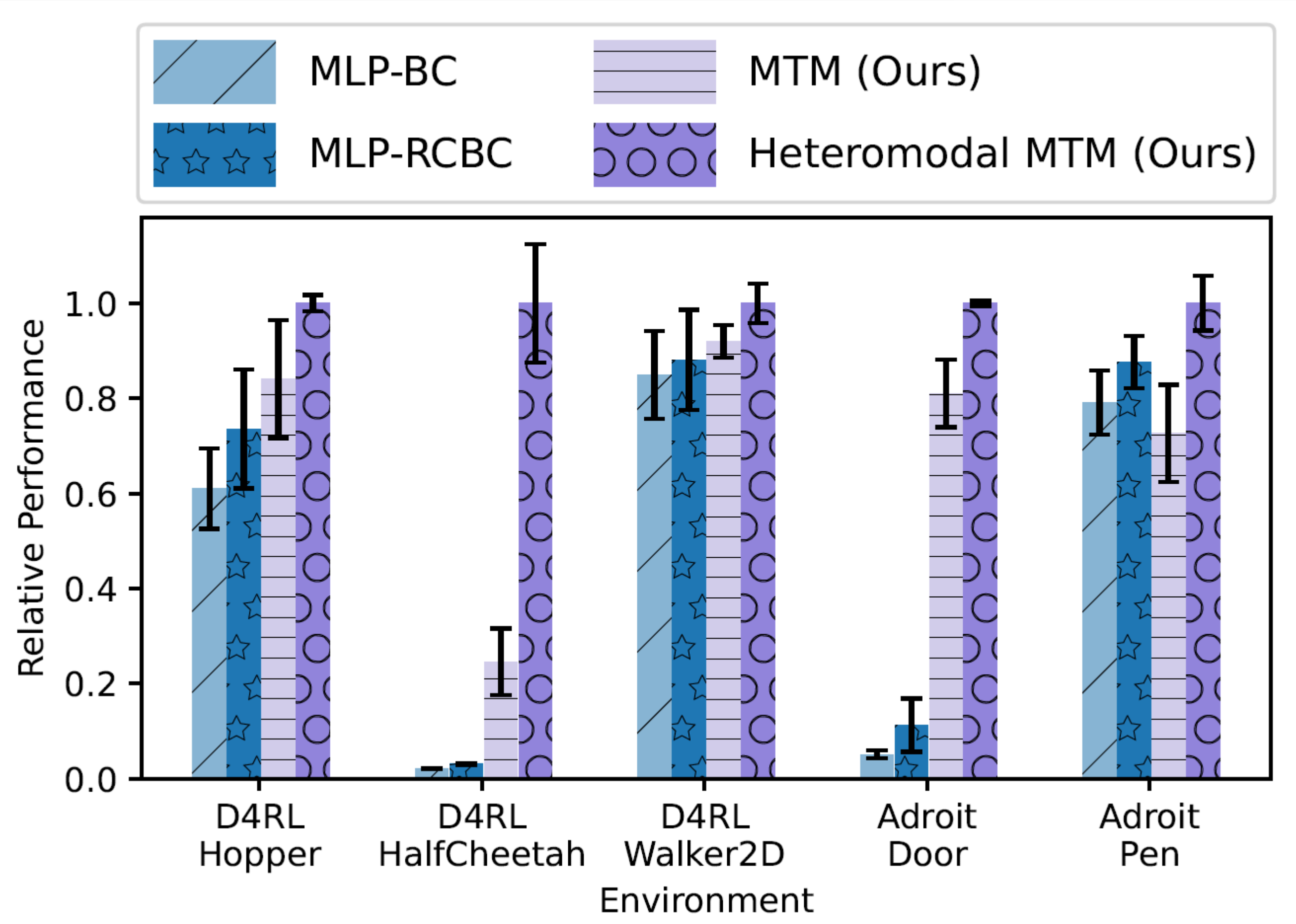
Impact of Masking Patterns



Autoregressive random often outperforms random, and in most cases is even competitive with the specialized (or oracle) RCBC mask.

Experiments

MTM can effectively learn from heteromodal datasets



Training a MTM models on Expert datasets across domains where only a small fraction of the data have action labels.

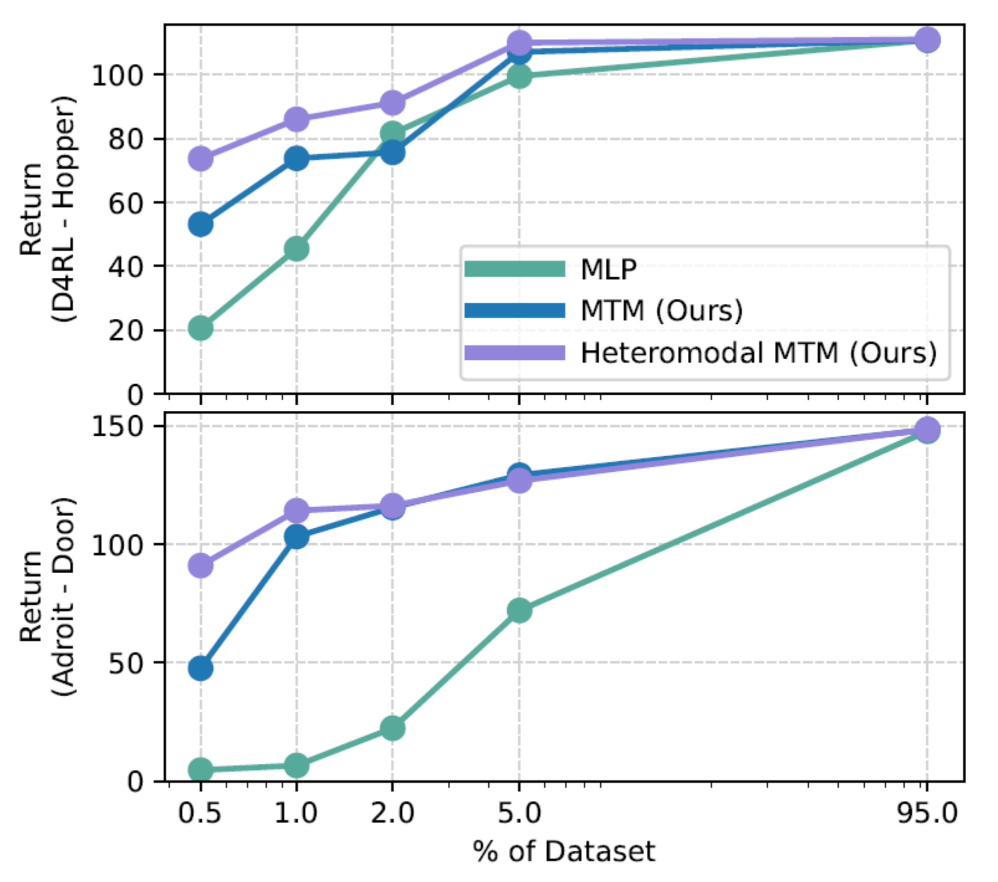
MLP and MTM: trained only on the subset of data with action labels.

Heteromodal MTM: two-stage action inference procedure.

- predict future states given current state and desired returns
- predict actions using the current state and predicted future states

Experiments

Dataset efficiency

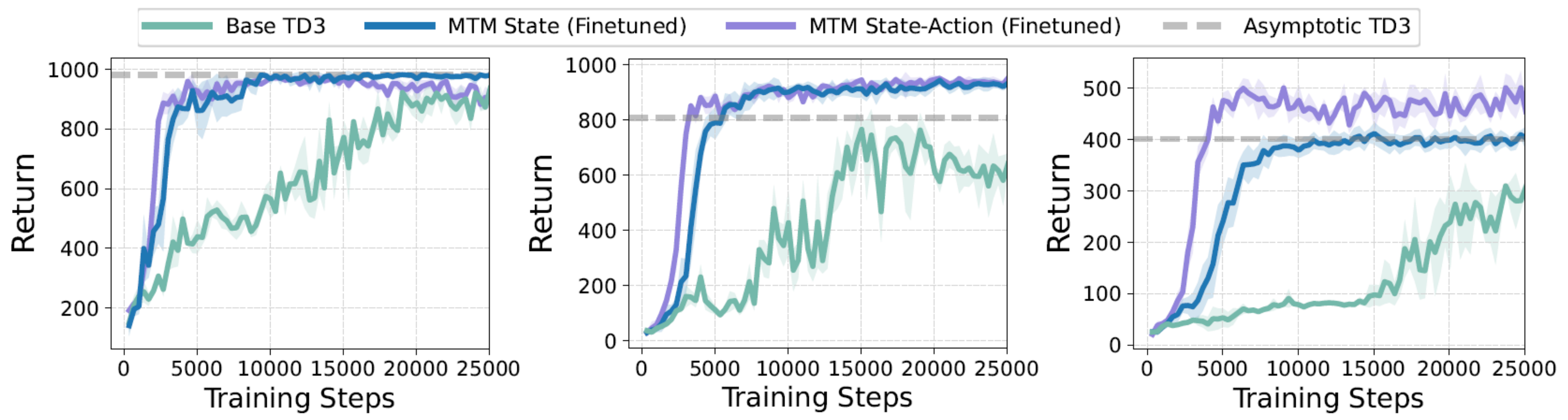


MTM is able to consistently outperform specialized MLP models in the low data regime.

Experiments

MTM Representations enable faster learning.

The agent is trained offline using data from the ExORL dataset.



(a) DM-Control Walker2D Stand Task. (b) DM-Control Walker2D Walk Task (c) DM-Control Walker2D Run Task

Base TD3: run TD3 on the dataset

MTM State: use state representations

MTM State-Action: additionally use state-action representations for critic

Finetuned: end to end finetuning of the representations during training

Conclusion

This paper introduced MTM as a versatile and effective approach for sequential decision making.

A single pretrained model can be used for different downstream purposes:

- inverse dynamics
- forward dynamics
- imitation learning
- offline RL
- representation learning

by simply changing the masks used at inference time.

THANKS!